Want to measure the path integral of free partical(no potential energy) from x_0 to x_f, which is to measure:

$$\int \mathrm{Dx}\,\Phi[x], \tag{1}$$

here $\Phi[x]$ is the function of path, Dx is a measure(explain latter).

1 Examine Dx

Assume the x-t trajecture of the partical is piece-wise linear, which means we can use linear function to interpolate the curve in a small time range. Set this partition to the curve $\{x_0, x_1, \ldots, x_N = x_f\}$,

We get:

$$\int_{x_0}^{x_f} \mathrm{Dx} = \int_{\mathbb{R}} \ldots \int_{\mathbb{R}} dx_1 \ldots \mathrm{dx}_{N-1}. \tag{2}$$

In math, I believe $\int_{\mathbb{R}} \mathrm{dx}_1$ means at time t_1, the summation of prob. the partical gets to the range of dx_1. eg: particle at time t_1 is in the range of 3-3.1, where dx_1 is 3.1-3.

In this way, we split thie abstract measure into a product measure of $\mathbb{R}^{N-1}$ space.

2 Examine $\Phi[x]$

Consider the summation energy of particle, which is:

$$S[x] = \int_{t_0}^{t_f} \mathrm{dt} \left\{ \frac{1}{2} m \left(\frac{\mathrm{dx}}{\mathrm{dt}} \right)^2 - U(x(t)) \right\}. \tag{3}$$

Since it is free particle, $U(x) = 0$. By my knowledge, the prob. of a particle exists in some place is related to its energy, so here we define:

$$\Phi[x] = e^{\hat{i}\frac{S[x]}{\hbar}}. \tag{4}$$

3 Exact Computation

Since we assurm piece-wise linear, which means $x_i - x_{i+1}$ the particle is in constant velosity, hence (3) can be simplified to:

$$S[x] = \frac{m}{2\Delta t}((x_1 - x_0)^2 + \cdots + (x_f - x_{N-1})^2). \tag{5}$$

Combine (2),(5), if Fubini's theorem is workable, we can interchange the integral to get the product of $N-1$ integral:

$$\int_{x_0}^{x_f} \mathrm{Dx}\,\Phi[x] = \prod_{i \in [N-1]} \int_{\mathbb{R}} \mathrm{dx}_i \frac{m}{2\Delta t}(x_i - x_{i-1})^2 \times \frac{m}{2\Delta t}(x_f - x_{N-1})^2. \tag{6}$$

Not hard to find each integral is a Gaussian Integral, which solves the problem.

Remark 1. From the result of Gaussian Integral we may have the intuition that we are actually measuring each linear path of particle under Gaussian measure, where I believe in the theory of Gaussian process we have the similar way to do so. Anyway, we successfully transverse a functional integral into a integral on Euclidean space!